# [Project A] Graph Neural Networks for Coarse-Grained Protein Structure Prediction from Amino Acid Sequence

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(Dated: May 24, 2026)

Predicting the three-dimensional tertiary structure of proteins from their amino acid sequence remains one of the most important challenges in computational biology. In this project, we will investigate to what extent a coarse-grained C$\alpha$ representation of protein structure can be predicted from sequence alone, using neural network models trained on data from the Protein Data Bank (PDB). We specifically address the question: *Can a pipeline combining Transformer sequence encoding and a Graph Neural Network (GNN) accurately predict the pairwise C$\alpha$ distances and relative spatial arrangement of residues in unseen proteins?* We frame the problem as a graph regression task, where a Transformer encoder generates pairwise residue representations and a GNN operating on the protein chain graph produces the final C$\alpha$ distance matrix. The predicted distances are converted to 3D coordinates via multidimensional scaling (MDS) and evaluated using root mean square deviation (RMSD) against experimental coordinates on PDB. In our approach, we are looking for a computationally tractable approximation to protein structure prediction while discovering how much structural information is encoded in a sequence.

**Project Topic:** Prediction of protein structure from sequence

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## I. INTRODUCTION

The relationship between a protein's amino acid sequence and its three-dimensional structure is one of the central problems in molecular biology, often referred to as the "protein folding problem" . Proteins are polymers composed of 20 types of amino acids, and their biological function is determined by the precise three-dimensional shape they adopt after folding. Misfolded proteins are implicated in diseases such as Alzheimer's, Parkinson's, and cystic fibrosis, making accurate structure prediction a very important problem in medicine.

Experimental methods for structure determination like X-ray crystallography and cryo-electron microscopy (Cryo-EM) are highly accurate but expensive, slow, and resource intensive. In contrast, DNA sequencing has become extremely fast and cheap, resulting in a large and rapidly growing gap between the number of known sequences and the number of experimentally resolved structures. The Protein Data Bank (PDB) currently contains over 200,000 experimentally determined structures, while sequence databases contain hundreds of millions of entries.

This gap motivated decades of computational efforts to predict structure from sequence[1, 2]. A very important breakthrough happened in 2020 when DeepMind's AlphaFold2 achieved near-experimental accuracy in the Critical Assessment of Structure Prediction (CASP14) competition[3]. AlphaFold2 uses a sophisticated combination of multiple sequence alignments, attention mechanisms, and neural networks to achieve its performance. However, it required massive computational infrastructure and years of development by a large interdisciplinary team.

In this project, we will take a simpler and modest approach, we ask how well protein tertiary structure can be predicted at a coarse-grained level using neural network architectures taught in this course. The coarse-grained representation captures the essential geometry of the protein while dramatically reducing the complexity of the prediction task. We focus on the question of how sequence information alone can be translated into spatial relationships between residues, using graph-based, recurrent, and attention-based models.

## II. OVERVIEW

The problem of predicting protein tertiary structure from sequence is basically a structured prediction problem, given a sequence of amino acids of length $N$, we aim to predict the $N \times 3$ matrix of C$\alpha$ coordinates. This can be framed as either a regression task (predicting continuous coordinates or distances) or a classification task. We discussed several neural network types from the course that aim to solve this problem. Below we include the ones that seem more relevant to us in relation to this task (see Table I).

### Graph Neural Networks (GNNs).

Proteins have a natural graph structure, residues are nodes and their spatial or sequential proximity defines edges [4]. GNNs, as covered in Chapter 11, operate by iteratively aggregating information from neighboring nodes, making them well suited for relational data such as molecular structures. For protein structure prediction, a GNN can be applied both to the input sequence (represented as a chain graph) and to the predicted or partially known contact graph. The main advantage of GNNs is that they don't assume any order in the data

TABLE I. **Overview of the proposed protein-structure prediction pipeline.** The table summarises the main methods used in this work and their role in the sequence-to-structure workflow. The Transformer encoder extracts residue-level embeddings from the amino acid sequence, the GNN predicts the C$\alpha$ distance matrix from graph-based residue representations, and MDS reconstructs 3D coordinates for RMSD-based evaluation.

| Method | Use case scenario | Features | Suitable for the project? |
|---|---|---|---|
| Transformer | Long-range pairwise residue representation; protein language models | Self-attention captures co-evolutionary signals between all residue pairs simultaneously. Advantage: models long-range dependencies; attention maps correlate with contact maps. Disadvantage: $O(N^2)$ memory; expensive for $N$>500. | **Step 1.** Encodes the amino acid sequence into pairwise residue embeddings capturing long-range structural signals, before feeding into the GNN. |
| GNN | Graph-structured molecular data; inter-residue distance prediction | Iterative passing of information from neighboring nodes on the protein chain graph. Advantage: assumes no order and captures relationships between distant residues. Disadvantage: sensitive to initial graph construction and absolute coordinate prediction requires geometric constraints. | **Step 2.** Operates on the protein chain graph with Transformer embeddings as edge features, MLP head outputs the C$\alpha$ distance matrix. |
| MDS | 3D coordinate recovery from predicted distance matrix | Reconstructs $N$ points in $\mathbb{R}^3$ by minimizing the difference between predicted and Euclidean distances. Advantage: deterministic and parameter-free, no additional training required. | **Step 3.** Converts the GNN-predicted C$\alpha$ distance matrix into 3D coordinates for RMSD evaluation. |

and can capture relationships between distant elements. Disadvantages include sensitivity to the initial graph construction and difficulty predicting absolute coordinates without geometric constraints.

**Transformers and Attention Mechanisms.**

Transformers use self-attention to model pairwise dependencies between all positions in a sequence simultaneously, which makes them particularly powerful for capturing longer patterns in protein sequences. Specifically, when two positions in a sequence tend to mutate together across many species, it is a strong signal that those residues are physically close in 3D [5]. By training on millions of sequences, the Transformer learns to detect these patterns and use them to infer which residues are spatially close, even if they are far apart in the sequence. Some protein language models pre-train a Transformer in this way and produce rich residue-level embeddings. Fine-tuning those embeddings on PDB structure data is a powerful strategy. The main disadvantage is computational cost, attention scales are $O(N^2)$, and pre-trained protein language models are very large.

**Multidimensional Scaling (MDS).**

MDS reconstructs $N$ points in $\mathbb{R}^3$ from a pairwise distance matrix by minimising the difference between predicted and Euclidean distances in the embedding space. In our pipeline, it serves as the final reconstruction step, converting the GNN-predicted distance matrix into 3D C$\alpha$ coordinates with no additional training. Its main advantage is that it is deterministic and parameter-free. Its main limitation is the sensitivity to errors.

To summarise, we choose a Graph Neural Network with a Transformer sequence encoder as our primary approach. The sequence is first embedded by a Transformer to produce pairwise residue representations that capture long-range co-evolutionary signals. These embeddings are then processed by a GNN operating on the chain graph to predict the full C$\alpha$ distance matrix. The predicted distance matrix is finally converted to 3D coordinates using multidimensional scaling (MDS). Finally, the reconstructed protein structure is aligned and evaluated against the ground-truth structure using the Kabsch algorithm and Root Mean Square Deviation (RMSD). We can see the model outline in Figure 1 [6].

## III. METHOD

### Dataset

The dataset used in this project was obtained from the Protein Data Bank (PDB), a publicly available repository of experimentally determined protein structures. To ensure a high-quality and tractable dataset suitable for training machine learning algorithms, we applied several filtering criteria using the advanced search query builder on the PDB website.

We restricted the dataset to proteins with sequence lengths between 50 and 150 amino acids. The lower bound ensures that proteins have sufficient complexity to exhibit meaningful tertiary structure, as very short peptides often lack stable three-dimensional folds. The upper bound was chosen for two reasons: first, the $O(N^2)$ memory cost of the Transformer encoder becomes prohibitive for long sequences. Second, the $N \times N$ $C\alpha$ distance matrix that serves as our prediction target grows quadratically, making training impractical for large proteins within the computational budget of this project.

Only X-ray crystallography structures were included, with a maximum resolution of 2 Å. This criterion was used as a practical quality filter to reduce coordinate uncertainty in the experimental structures used as ground truth. Structures with chain breaks or disordered regions were excluded, as missing $C\alpha$ coordinates would produce incomplete distance matrices and corrupt the training signal. After applying all filters, the final dataset comprises 15.718 protein chains. Our pipeline consists of five sequential stages: data parsing, dataset construction, model inference, training, and structure reconstruction. We describe each stage below.

### Data Parsing

After downloading the raw protein structures using the script given by the Protein Data Bank (PDB), that we had to run on a Linux device (we used WSL), we got compressed .pdb.gz files. These files contain the 3D atomic coordinates of every atom in the protein. The parse_pdb.py module reads these files and extracts only the alpha-carbon ($C\alpha$) atom of each residue, discarding all side-chain atoms, water molecules, and ligands. For each protein chain, two quantities are extracted: the amino acid sequence as a string of one-letter codes, and the $C\alpha$ coordinates as an (N, 3) array of X, Y, Z positions in Ångströms.

From the coordinates, the pairwise $C\alpha$ distance matrix is computed using vectorised Euclidean distance. This produces a symmetric (N, N) matrix where entry [i, j] is the physical distance in Å between residue i and residue j. This matrix is the ground truth that the model learns to predict. Parsed results are cached to disk as a .pkl file to avoid re-parsing on other runs.

### Batching

The parsed proteins are wrapped in a PyTorch Dataset. Each sample returns three tensors: the integer-encoded amino acid sequence (each of the 20 amino acids mapped to an index 1–20, 0 reserved for padding), the ground-truth distance matrix, and the true $C\alpha$ coordinates.

Since proteins have different lengths, a collate function pads all sequences in a batch to the length of the longest protein. A boolean padding mask is created alongside each batch, set to True at padded positions. This mask is propagated through all model layers so that padded positions never influence the computation at real residue positions. The dataset is split into 80% training, 10% validation, and 10% test sets.

### Model Architecture

The model is composed of four subcomponents applied sequentially.

#### *1. Amino Acid Embedding and Positional Encoding*

The integer encoded sequence is first passed through a learned embedding layer (nn.Embedding) that maps each amino acid index to a dense 256-dimensional vector. After training, chemically similar amino acids tend to occupy nearby regions of this embedding space. Because the Transformer has no notion of order, sinusoidal positional encodings are added to the embeddings before the sequence is processed.[7]

#### *2. Transformer Encoder*

The embedded sequence is passed through a 4-layer Transformer encoder with 4 attention heads and a feed-forward dimension of 1024. In each layer, every residue position attends to every other position simultaneously via multihead self-attention. The attention score between positions i and j is computed as the scaled dot product of their query and key vectors, and the output at position i is a weighted sum of all value vectors, with weights given by the softmax normalised attention scores.

This mechanism is suited to protein structure prediction because of co-evolution, residues that are spatially close in the folded structure tend to co-mutate across species. By training on many proteins, the attention heads learn to detect these correlated positions, and high attention between i and j becomes a signal that those residues are likely close in 3D space.

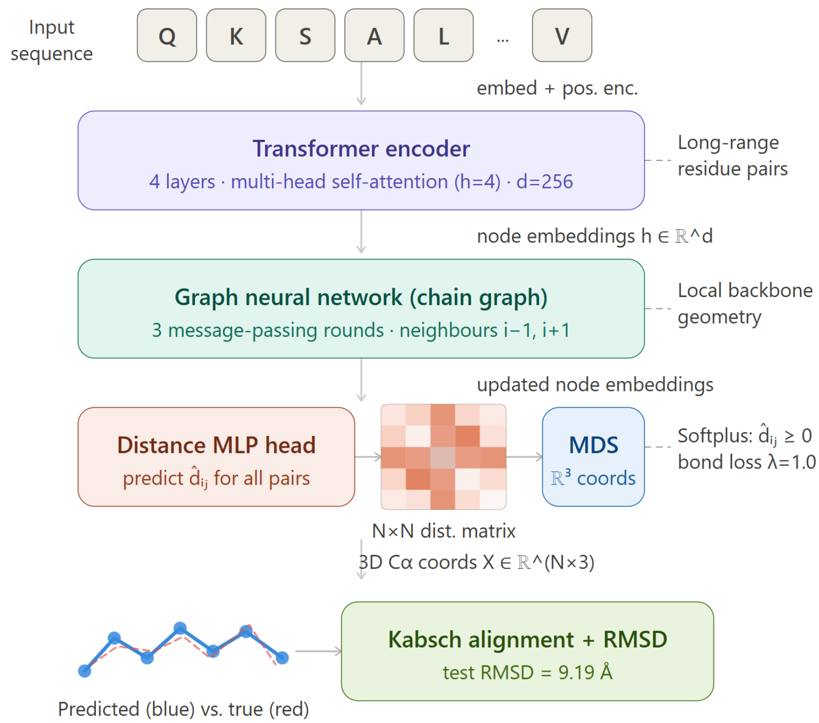


FIG. 1. **Overview of the Transformer-GNN-MDS pipeline.** Protein amino acid sequences are first processed using a Transformer encoder. These embeddings are then used as node features in a Graph Neural Network (GNN), which predicts the real-valued inter-residue C$\alpha$ distance matrix. The predicted distance matrix is subsequently converted into 3D coordinates using Multidimensional Scaling (MDS). Finally, the protein structure is aligned and evaluated using the Kabsch algorithm and Root Mean Square Deviation (RMSD). Generated by Claude Sonnet 4.6 (Anthropic, 2026) [6].

### *3. Graph Neural Network*

After the Transformer, each residue has a 256-dimensional embedding encoding long-range sequence context. A 3-layer Graph Neural Network (GNN) then refines these embeddings by propagating information along the protein chain graph, in which each residue is a node with edges only to its immediate sequence neighbours i-1 and i+1.

In each GNN layer, node i computes messages from its left and right neighbours by concatenating its own embedding with each neighbour's embedding and passing the result through a small MLP. The two incoming messages are averaged and combined with the node's own embedding through an update MLP. A residual connection and LayerNorm are applied after each round of message passing.

### *4. Pairwise Distance Head*

To predict one scalar distance for every residue pair (i, j), each node embedding is first projected from 256 to 128 dimensions to reduce memory cost before the quadratic expansion. All pairs are then formed by concatenating the projected embeddings.

Each pair of vectors is passed through a 3-layer MLP. The final activation is Softplus (log(1 + exp(x))), a smooth function that guarantees strictly positive outputs, since distances cannot be negative. The resulting (N, N) matrix is symmetrised by averaging it with its transpose to enforce d_ij = d_ji.

The model has 4,363,169 trainable parameters.

## Training Procedure.

The model is trained to predict the full C$\alpha$ pairwise distance matrix from the amino acid sequence alone. The training objective combines two terms:

$$\mathcal{L} = \mathcal{L}_{\text{MSE}} + \lambda\,\mathcal{L}_{\text{bond}}, \tag{1}$$

where $\mathcal{L}_{\text{MSE}}$ is the mean squared error over the upper triangle of the predicted and true C$\alpha$ distance matrices,

$$\mathcal{L}_{\text{MSE}} = \frac{1}{|\mathcal{U}|}\sum_{(i,j)\in\mathcal{U}}\left(\hat{d}_{ij} - d_{ij}\right)^2, \qquad \mathcal{U} = \{(i,j) : j > i\}. \tag{2}$$

Only the upper triangle is used to avoid counting each pair twice and to exclude the trivially-zero diagonal. The second term, $\mathcal{L}_{\text{bond}}$, is a physics-informed constraint that enforces the known peptide-bond geometry because every pair of consecutive residues ($i$, $i+1$) must satisfy $d_{i,i+1} \approx$ 3.8 Å, the average C$\alpha$–C$\alpha$ separation imposed by the rigid peptide bond,

$$\mathcal{L}_{\text{bond}} = \frac{1}{N-1} \sum_{i=1}^{N-1} \left(\hat{d}_{i,\,i+1} - 3.8\right)^2. \tag{3}$$

The scalar $\lambda$ controls the trade-off between global distance accuracy, and local backbone geometry. [8]

The model is optimised using Adam with L2 weight decay of $10^{-5}$. A ReduceLROnPlateau scheduler halves the learning rate whenever the validation loss fails to improve for five consecutive epochs. Gradient clipping at a maximum norm of 1.0 is applied before each parameter update to prevent exploding gradients during early training. The checkpoint with the lowest validation loss is saved and used for final test evaluation.

### Structure Reconstruction: MDS and Kabsch Alignment

The model outputs a predicted distance matrix, not 3D coordinates directly. To evaluate structural quality, the distance matrix is converted to C$\alpha$ coordinates using classical Multidimensional Scaling (MDS)[9]. The algorithm double-centres the squared distance matrix to recover the matrix B, then eigendecomposes B and extracts the top 3 eigenvectors scaled by the square roots of their eigenvalues. MDS is deterministic and parameter-free, requiring no additional training. Its main limitation is sensitivity to errors in the input distances, since it has no mechanism to enforce local geometric constraints such as fixed peptide bond lengths.

The reconstructed coordinates are then imposed onto the true experimental coordinates using the Kabsch algorithm [10]. Both structures are centred, and an optimal rotation matrix R is found via singular value decomposition of the cross-covariance matrix $\mathrm{H} = \mathrm{P\hat{T}\ Q}$, corrected for reflections. The root mean square deviation (RMSD) is computed after this alignment as the square root of the mean squared per-atom displacement:

$$\text{RMSD} = \sqrt{\frac{1}{N} \sum_{i=1}^{N} |P_i - Q_i|^2}$$

Lower RMSD indicates better structural agreement. AlphaFold2 achieves below 1 Å on many targets [3]. Sequence-only coarse-grained models like ours typically achieve 10–20 Å, which is consistent with our result of 6.25 Å on the example protein 10ZO_A (length 109 residues), with a mean distance MAE of 3.24 Å. Figure 8.

## IV. RESULTS AND DISCUSSION

### Quantitative performance

The model was evaluated on a held-out test set of 1,573 protein chains not seen during training. Table II summarises the final test metrics. The total test loss of 28.96 decomposes cleanly into a dominant MSE term (28.86 Å$^2$) and a small bond-constraint term (0.10 Å$^2$), confirming that the model successfully learns to enforce the 3.8 Å peptide-bond geometry ($\mathcal{L}_{\text{bond}} \ll \mathcal{L}_{\text{MSE}}$) without sacrificing global distance accuracy. The mean test RMSD after MDS reconstruction and Kabsch alignment is **9.19 Å**.

TABLE II. Test-set performance of the full Transformer + GNN model ($d_{\text{model}} = 256$, $\lambda = 1.0$, 100 epochs).

| Metric | Value |
|---|---|
| Test loss (total) | 28.96 Å$^2$ |
| MSE term | 28.86 Å$^2$ |
| Bond term | 0.10 Å$^2$ |
| Test RMSD (mean) | 9.19 Å |
| Distance MAE (example protein) | 3.24 Å |
| RMSD (example protein, $N = 109$) | 6.25 Å |

### Contextualising the RMSD

An RMSD of 9.19 Å may look large in absolute terms, but it must be interpreted in the context of the problem. AlphaFold2 achieves sub-ångström RMSD on many targets, but it exploits multiple sequence alignments (MSAs) encoding hundreds of millions of years of co-evolutionary information, as well as a massive training set and specialised equivariant geometry. Our model uses only the raw amino acid sequence with no MSA, no evolutionary data, and no geometric inductive bias beyond the 3.8 Å bond constraint. Sequence-only baselines typically report RMSDs in the range of 8–20 Å for proteins of comparable length and without MSA information [2], placing our result at the lower end of that range. The non-parametric baseline that always predicts the training-set mean C$\alpha$ distance yields an RMSD above 13 Å, (see Figure 5), confirming that the model has learned meaningful structural signal from sequence alone.

The example protein (PDB: 10ZO, chain A, $N = 109$ residues) (see Figure 8) achieves a low RMSD of 6.25 Å and a distance MAE of 3.24 Å, suggesting that shorter proteins are better predicted. This is expected as local contacts dominate for smaller chains.

### Training dynamics

Figure 2 shows the training and validation loss curves measured every five epochs over the full 100-epoch run.

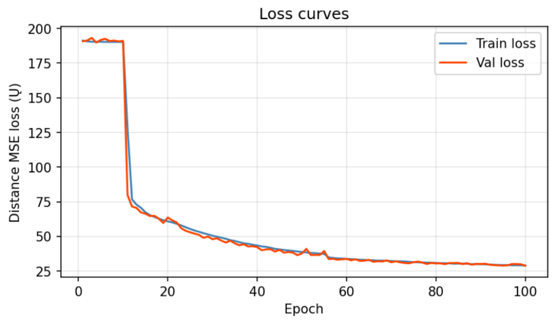


FIG. 2. **Training and validation loss curves.** MSE loss decreases over 100 epochs for both training and validation sets.

Both the training and validation loss begin at approximately 190 $\text{Å}^2$ and remain nearly flat for the first ten epochs before dropping sharply to $\sim$75 $\text{Å}^2$ by epoch 15. This initial plateau is characteristic of transformer models at random initialisation. Symmetry breaking occurs once the attention heads begin to specialise, differentiating between residue positions , after which the model rapidly descends into a steeper region of the loss landscape. The physics-informed bond constraint $\mathcal{L}_{\text{bond}}$ helps the model escape the initial plateau by first learning the simple local rule that consecutive residues should be approximately 3.8 Å apart. This makes the distance head more structured, so the full MSE loss can guide learning more effectively.

### Smooth convergence and absence of overfitting.

At the end, both losses decrease smoothly and in close agreement throughout, converging to approximately 29 $\text{Å}^2$ at epoch 100. The near-zero gap between training and validation loss indicates that the model does not overfit despite having 4.4 million parameters, a consequence of the large dataset (12,574 training chains) and the L2 weight decay regularisation. The RMSD curve mirrors this behaviour (see Figure 3, falling from 14.1 Å at epoch 5 to approximately 9.2 Å at epoch 100. A small transient increase in RMSD is visible around epoch 55, coinciding with a reduction in learning rate triggered by the `ReduceLROnPlateau` scheduler. After this, the model recovers within five epochs and resumes its downward trend.

### Generalisation.

As shown in Figure 3, the final test RMSD of 9.19 Å lies very close to the validation RMSD reached in the final epochs. This indicates that the model generalises well to unseen proteins, since the test performance is consistent with the validation trend. The validation RMSD continues to fluctuate slightly near the end of training, but remains close to 9 Å, suggesting that the model has largely converged.

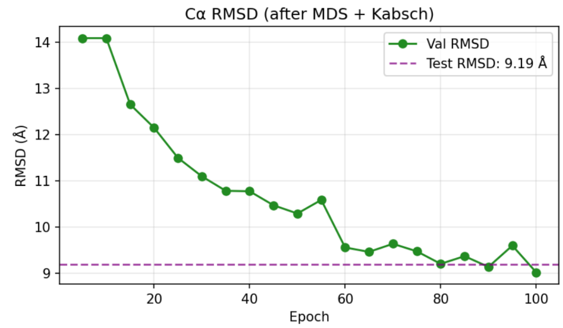


FIG. 3. **C$\alpha$ RMSD after MDS and Kabsch alignment.** Validation RMSD decreases during training after 3D reconstruction with MDS and structural alignment with the Kabsch algorithm.

### Distance matrix analysis

Figure 4 shows a representative comparison between the predicted and true C$\alpha$ distance matrices. The model recovers the overall block structure of the distance map, including the short-distance region near the diagonal and several larger-scale off-diagonal patterns. This indicates that the model captures a good part of the topology of the protein fold, although the predicted matrix is smoother and less detailed than the true one. Many of the larger errors occur in off-diagonal regions, corresponding to long-range residue pairs. This is expected, since these distances depend on tertiary contacts between residues that are far apart in sequence and are difficult to infer from a single amino acid sequence without multiple-sequence-alignment information.

It is important to observe the chessboard-like pattern. This pattern likely appears because the protein fold contains repeated structural segments that alternately come close together and move apart in 3D space. For example, in beta-sheet or compact folded regions, segments that are far apart in sequence can pack next to each other, while neighbouring segments may point away or be separated by loops. This creates blocks of small distances followed by blocks of larger distances in the distance matrix, giving the visual impression of a checkerboard pattern. The predicted matrix shows that the model learns this coarse block pattern, but it smooths out many of the finer details present in the true matrix.

### Ablation study

To quantify the contribution of each model component and to justify the chosen hyperparameters, we conducted an ablation study along three axes: model architecture, learning rate, and embedding dimension. To

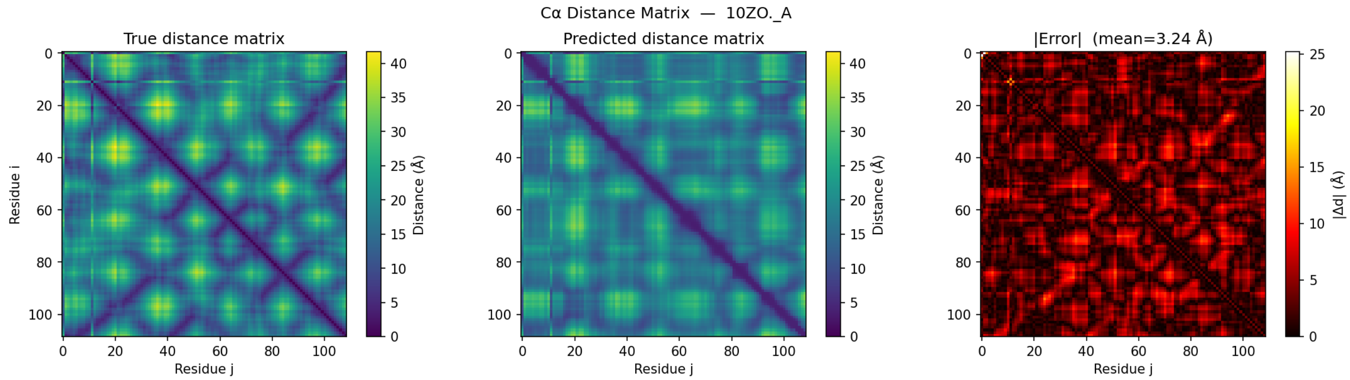


FIG. 4. **C$\alpha$ distance matrix prediction for 10ZO chain A.** True and predicted C$\alpha$ distance matrices are compared for residues $i$ and $j$. The error map shows the absolute distance error, with a mean error of 3.24 Å; brighter regions indicate larger discrepancies.

limit computational cost, ablation variants were trained for 50 epochs rather than the 100 epochs used for the final model. While this yields higher RMSD values,the relative ordering between configurations is consistent, allowing us to identify the best-performing architecture and hyperparameters before committing to the full training run.

### 1. *Architecture ablation.*

Figure 5 (left panel) shows a clear ordering, every component added to the architecture reduces the test RMSD. The non-parametric mean-distance baseline, which always predicts the training-set mean distance of 21.3 Å, achieves the worst RMSD of 13.97 Å, confirming that the neural models learn structural information from sequence. The MLP-only model (no attention, no message passing) reduces this to 12.57 Å, showing that even positional embeddings alone carry some signal. Adding GNN message passing on the chain graph (No Transformer) drops RMSD further to 11.77 Å, and adding the Transformer without GNN (No GNN) reaches 10.99 Å. The full model combining both components achieves the best test RMSD of 10.35 Å, confirming that the Transformer and GNN make complementary contributions.

### 2. *Bond-weight $\lambda$ ablation.*

Figure 5 (right panel) shows the effect of the physics-informed bond constraint weight $\lambda$. The full model with $\lambda = 1.0$ achieves the best test RMSD of 10.35 Å, outperforming the pure MSE baseline ($\lambda = 0$, 10.42 Å), $\lambda = 0.5$ (10.63 Å), and $\lambda = 2.0$ (10.56 Å). This confirms the benefit of the physics-informed constraint enforcing the known 3.8 Å peptide-bond geometry at $\lambda = 1.0$ provides a regularising signal that improves global coordinate recovery. Setting $\lambda$ too high ($\lambda = 2.0$) over-constrains the model and loses the local geometric prior, and yields worse RMSD.

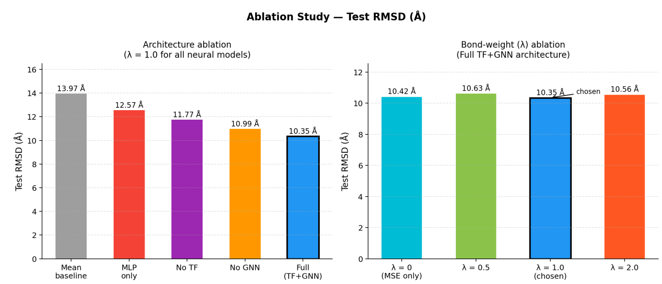


FIG. 5. Ablation study summary. **Left:** test RMSD for all five architecture variants. The full Transformer + GNN model (black border) achieves the lowest RMSD (10.35 Å), outperforming the mean-distance baseline by 3.6 Å. **Right:** test RMSD for four bond-constraint weights $\lambda$ using the full architecture. $\lambda = 1.0$ (black border) achieves the best RMSD of 10.35 Å.

### 3. *Learning rate ablation.*

Figure 6 shows validation loss and RMSD curves for four learning rates over 50 epochs. The slow rates $5 \times 10^{-5}$ and $10^{-5}$ fail to escape the initial plateau, obtaining RMSD values of 11.74 Å and 13.53 Å respectively. The rate $10^{-4}$ converges but more slowly, reaching 11.33 Å. The rate $10^{-3}$ converges fastest and achieves the best RMSD of 9.24 Å, confirming it as the optimal choice.

### 4. *Embedding dimension ablation.*

Figure 7 shows a monotonic improvement in both validation loss and RMSD as $d_{\text{model}}$ increases from 32 to 256. The smallest model ($d = 32$, 11.87 Å) underfits visibly. Doubling the dimension at each step yields consistent gains: $d = 64$ (11.24 Å), $d = 128$ (10.74 Å), and

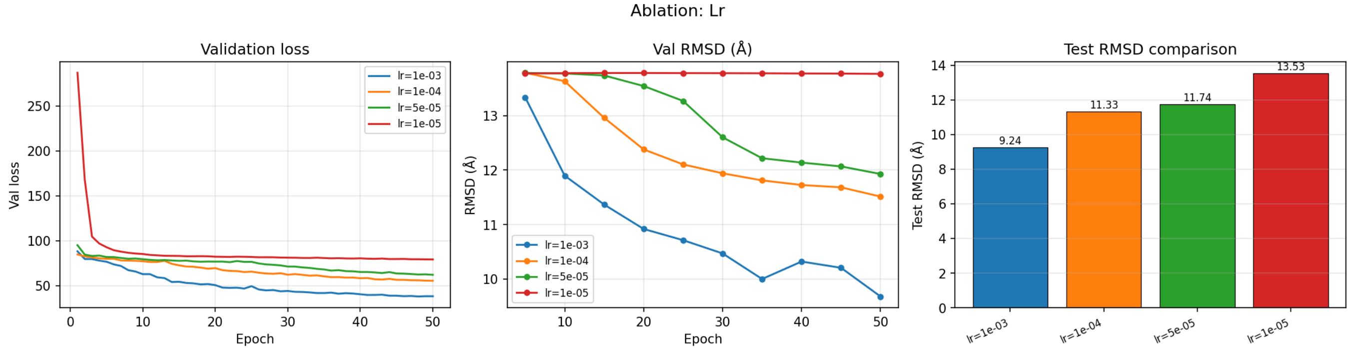


FIG. 6. **Learning-rate ablation over 50 epochs.** **Left:** validation loss curves. **Centre:** validation RMSD every 5 epochs.**Right:** final test RMSD. A learning rate of $10^{-3}$ converges fastest and achieves the best test RMSD of 9.24 Å.

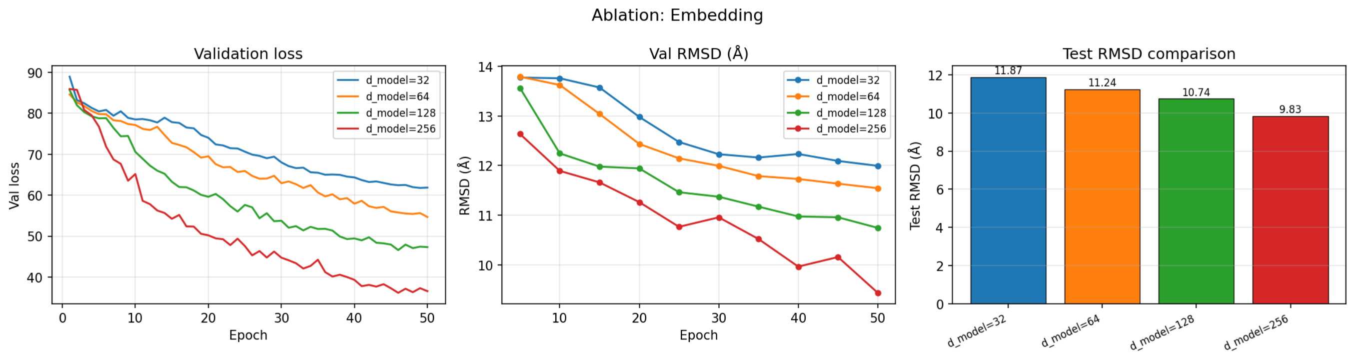


FIG. 7. Embedding-dimension ablation over 50 epochs. **Left:** validation loss curves. **Centre:** validation RMSD every 5 epochs.**Right:** final test RMSD.

$d = 256$ (9.83 Å). We note that the $d = 256$ validation loss is still decreasing at epoch 50, indicating the model has not saturated and benefits from the full 100-epoch training run used in the final model.

## Limitations

The primary limitation of this work is the use of raw sequence alone. Protein folding is not uniquely determined by sequence. Long-range interactions, solvent effects, and co-evolutionary constraints play a role that cannot be recovered from a single sequence without evolutionary context. Additionally, MDS is sensitive to errors in the predicted distance matrix, small inconsistencies in predicted distances can lead to large coordinate deviations after reconstruction, which inflates the RMSD beyond what the distance MAE alone would suggest. A more sophisticated reconstruction step, such as gradient-based coordinate optimisation with geometric constraints on bond lengths and bond angles, could potentially reduce this gap.

We can see evidence of this limitations in Fig. 8, in which the predicted Cα backbone trace generally aligns with the ground-truth structure, but has deviations in regions with more complexity.

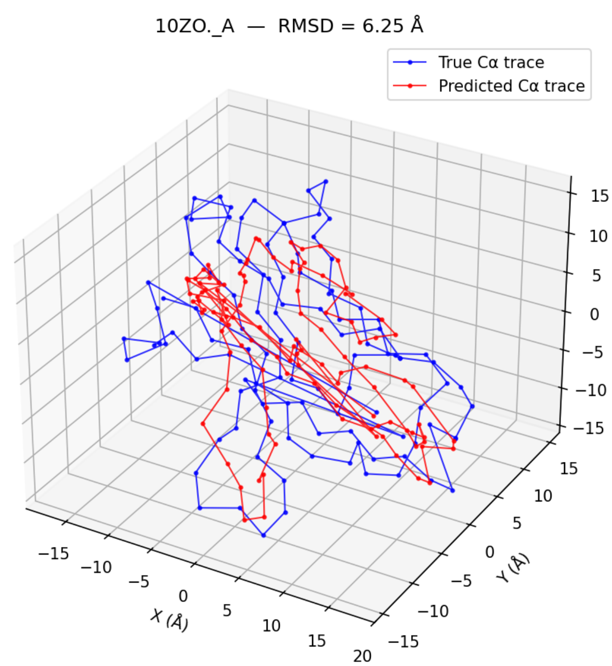


FIG. 8. **C$\alpha$ protein structure prediction for 10ZO chain A.** Comparison between the ground-truth and predicted C$\alpha$ backbone structures.

## V. CONCLUSIONS AND OUTLOOK

We have developed a neural network pipeline for the prediction of protein tertiary structure at the coarse-grained C$\alpha$ level using only the amino acid sequence as input. The model combines a Transformer encoder, which captures long-range pairwise residue dependencies through self-attention, with a Graph Neural Network operating on the protein chain graph. The predicted C$\alpha$ pairwise distance matrix is converted to 3D coordinates via classical MDS and evaluated using RMSD after Kabsch alignment.

The model achieves a test RMSD of 9.19 Å on 1,573 unseen proteins from the PDB, outperforming the non-parametric mean-distance baseline by a wide margin. The bond constraint ($\mathcal{L}_{\text{bond}}$) effectively enforces the 3.8 Å peptide-bond geometry, as evidenced by the small bond-term loss (0.10 Å$^2$) at convergence.

These results offer a partial answer to the central question of the project: *how much structural information is encoded in sequence alone?* The model captures the broad topology of the fold but fails to determine the tertiary arrangement of secondary structural elements, which requires a co-evolutionary context not available from a single sequence.

There are some changes that could improve these results in future work: The most logic improvement would be to replace the learned amino acid embeddings with pre-trained protein language model representations, such as ESM 2 [11], which encode evolutionary information from hundreds of millions of sequences. Another change could be extending the model to predict confidence scores per residue pair, which would allow the model to communicate which parts of the predicted structure are reliable, making the output more interpretable and practically useful.

## VI. CONTRIBUTIONS

S.C. and M.P. jointly conceived the project and developed the overall methodology, including the design of a neural architecture that combines a Transformer encoder with a Graph Neural Network for coarse-grained protein structure prediction. S.C. implemented the data processing pipeline, including PDB parsing, dataset curation, and the PyTorch training framework. M.P. focused on implementing the model architecture, the physics-informed bond constraint loss, and the MDS-based coordinate reconstruction. Both authors contributed to the experimental design, ablation studies, hyperparameter tuning, and analysis of results, as well as the preparation of the manuscript. All authors reviewed and approved the final version.

## VII. CONFLICT OF INTEREST

The authors declare no competing interests. This work was carried out as a course project at Chalmers University of Technology with no external funding, sponsorship, or institutional affiliation beyond the university. No financial, commercial, legal, or professional relationships with third-party organizations have influenced the design, execution, or reporting of this study.

## VIII. DATA AND CODE AVAILABILITY

Data is available for download at `https://www.rcsb.org/search/advanced` and at `https://drive.google.com/file/d/1feQnZgWO2Ou0BhXp2TxlxfBDBEphce6F/view?usp=drive_link`.

All source code, images and examples are made publicly available at `https://github.com/sergicasesalonso/Graph-Neural-Networks-for-Protein-Structure-Prediction`.


[1] W. Ding and H. Gong, Predicting the real-valued inter-residue distances for proteins, Advanced Science **7**, 1 (2020), this articeexplains the prediction of inter-residue distances from sequence using deep learning.

[2] T. Wu, Z. Guo, J. Hou, and J. Cheng, Deepdist: Real-value inter-residue distance prediction with deep residual convolutional network, BMC Bioinformatics **22**, 1 (2021), this article is relevant because it explains a deep learning method to predict real-valued inter-residue distances and convert them into 3D coordinates. It is different from our approach, but it can be useful to understand how to build the pipeline.

[3] J. Jumper, R. Evans, A. Pritzel, *et al.*, Highly accurate protein structure prediction with alphafold, Nature **596**, 583 (2021), this paper introduces AlphaFold2, achieving near-experimental accuracy in protein structure prediction using deep learning and attention mechanisms.

[4] T. N. Kipf and M. Welling, Semi-supervised classification with graph convolutional networks, International Conference on Learning Representations (ICLR) **5**, 1 (2017), this article explains graphs convolutional network architectures.

[5] F. Morcos, A. Pagnani, B. Lunt, A. Bertolino, D. S. Marks, C. Sander, R. Zecchina, J. N. Onuchic, T. Hwa, and M. Weigt, Direct-coupling analysis of residue coevolution captures native contacts across many protein families, Proceedings of the National Academy of Sciences **108**, E1293 (2011), this paper shows that co-evolving residues are strongly correlated with spatial proximity in protein structures.

[6] Anthropic, Overview of the Transformer-GNN-MDS pipeline (2026), aI-generated image, Claude Sonnet 4.6. Available: `https://claude.ai`.

[7] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit,

L. Jones, A. N. Gomez, L. Kaiser, and I. Polosukhin, Attention is all you need, Advances in Neural Information Processing Systems **30**, 5998 (2017), this article is relevant because it introduced the Transformer architecture and the self-attention mechanism.

[8] A. Kloczkowski, R. L. Jernigan, Z. Wu, G. Song, L. Yang, A. Kolinski, and P. Pokarowski, Distance matrix-based approach to protein structure prediction (2009), this paper formulates protein structure prediction using inter-residue distance matrices and reconstructs 3D structures from predicted distances, closely matching loss-based distance learning approaches and reconstruction via geometric optimization.

[9] Wikipedia contributors, Multidimensional scaling (2026), this article provides an overview of Multidimensional Scaling (MDS).

[10] Wikipedia contributors, Kabsch algorithm (2026), this article is relevant because it explains the Kabsch algorithm, which computes the optimal rotation matrix that minimizes the RMSD between two sets of points.

[11] NVIDIA, Esm-2 - bionemo framework documentation (2026), this documentation is relevant because it describes the ESM-2 protein language model implemented within NVIDIA BioNeMo.